



MIT SLOAN SCHOOL OF MANAGEMENT
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MODULE 1 UNIT 2
Media Set Video 3 Transcript

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Adding steps

JACOB ANDREAS: So far, everything we've asked this agent to do is pretty easy. What if we wanted to give it a more complicated task? For example, if our company return policy said that customers were entitled to free shipping, if any two of the following three conditions held, the next month had five Thursdays, the moon was in a waxing gibbous phase, or the order number was co-prime with 255.

Now, an important feature of even the most powerful language models that exist today is that they can't always generate answers to questions immediately. Sometimes they need to think out loud. For example, we've given the language model the instruction to compute the cube root of some complicated number immediately, and it responds and it comes up with some answer. And it turns out that this answer is not quite right. The cube root is close to 233, but not exactly 233.

We've given the language model almost exactly the same instruction, but we've told it to give step-by-step instructions or to think step by step before returning an answer. And what we see now is that the language model produces a longer and much more complicated answer. And in the course of coming up with that longer and more complicated answer, it's actually able to produce also a more reliable answer that's accurate to a greater number of decimal places.

If you've heard of chain of thought reasoning in language models, this is an example of this. We simply instruct the language model to think step by step, and this allows it to generate more computation in the course of coming up with an answer, and usually as a result to produce better answers. This is true not just for sort of hard math or formal reasoning problems, but also situations where language models need to combine information from multiple sources, reason about the far future consequences of their actions, or solve challenging constraint satisfaction problems.

The main takeaway here is that we can take a language model agent and make it more capable or more able to reason about the long-term consequences of its actions by separating out its inner monologue, which it uses for figuring out what to do, from its external dialogue, which it uses either to interact with tools or for communicating with the user.

So, for example, when the language model sees this complicated shipping policy as a result of one of its tool calls, it can think step by step about whether the user's current situation applies to this shipping policy before actually responding to the user. And again, we can implement this either just by instructing the language model to do some step-by-step thinking or by giving it an extra tool call, which does nothing but which allows it to sort of wrap its thoughts in a bit of code that will hide it from the rest of the interaction.

Another thing that we can do here is use code to reason and not just to take actions. So when we're thinking about this shipping policy, there's a lot of math that we have to do involving dates of months, there's a lot of knowledge that we need to know about what days particular days of the week are going to fall on. And rather than requiring the language model to be able to reason about this in natural language, or to have memorized all of

these facts and stored them in its internal weights, we can simply give the language model the ability to write code that calls sort of utility functions for processing dates, and use this to offload some of the reasoning that it does for figuring out whether the shipping policy applies.

Now, this kind of planning is a feature of language models in general and not just language model agents. But there's one other trick that we can use when working with agents specifically, and that's explicitly giving these agents the ability to plan before they act. For example, in the conversation that we've been looking at so far, it's maybe not immediately obvious what parts of the company knowledge base the language model needs to search for when figuring out how to handle the user's particular situation.

On one hand, you might imagine that the language model needs to do a general search for a return policy, but that what comes back doesn't actually say anything about gifts and that there's a separate article about gifts. On the other hand, you could imagine that just searching for gift return policy would give you the right article right away without having to go through these sequences of steps.

Simulating future outcomes

ANDREAS: So here's what we can do. We can allow the agent to imagine the consequences of each of these actions by trying out a couple of different ways of solving the user's problem in parallel, seeing what kind of state it got into, what information it had access to, and whether it was able to actually respond to the user's request or not. And only at the end of trying a couple sequences of possible actions, actually deciding which possible outcome or which final state of the world was the best one, and then doing whatever sequence of actions led to that outcome and not any of these others. So think of this as the agent imagining possible future states of the world and rolling the world forward in its imagination a few steps into the future, imagining what a copy of the agent might have done in each of those futures in order to figure out what action was best.

Now it's important to emphasize that this works only when you have access to a simulator for the world, and maybe even a simulator for the user that you're trying to assist, or the ability to undo any actions that you took. So it works great, for example, if we want to try a couple of different search strategies, or try interacting with websites or databases where we know we can undo whatever it was that we did, or where we have access to a simulator.

But if we want to explore possible futures that involve things like sending emails or charging credit cards, we're going to need to make sure that we can run these simulation procedures without actually affecting anything in the real world, but still being able to reason about their eventual consequences. So what we've done now is we've taken our basic language model agent architecture, and we've extended it with the ability to plan, either just by sort of talking out loud to itself in natural language or by imagining outcomes of interactions with users in simulated worlds.

SPEAKER: Did you understand all the concepts covered in this video? If you'd like to review any of the content, please click on the chapter menu.